Qualitativeness, Haecceitism, and the Identity of Indiscernibles

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1. Three PIIs

Trivial PII:If a has every property b has, a=b.

PII:If a has every **qualitative** property b has, a=b. ⇐ *our focus today*

Intrinsic PII: if a has every **intrinsic qualitative** property b has, a=b.

- Terminology: for any object x, the *haecceity* of x (what Adams calls its *thisness*) is the property of identical to x ($\lambda y.y{=}x$).

2. Qualitativeness

Intuitive characterization:

> We might try to capture the idea by saying that a property is purely qualitative—a suchness—if and only if it could be expressed, in a language sufficiently rich, without the aid of such referential devices as proper names, proper adjectives and verbs (such as 'Leibnizian' and 'pegasizes'), indexical expressions, and referential uses of definite descriptions. (PTPI 7)

Not just properties but relations and propositions can be qualitative.

A basic principle: if Y can be defined in terms of $X_1,\ldots,X_n$, and each of $X_1,\ldots,X_n$ is qualitative, then Y is qualitative.[1]

To avoid bringing in the somewhat difficult ideology of 'definability' you can use a schema:

Inheritance**:** if each of $X_1,\ldots,X_n$ is qualitative, then Φ is qualitative — where Φ is any term with no nonlogical constants and whose only free variables are among $X_1,\ldots,X_n$.

- The converse of this is obviously hopeless unless we require each of $X_1,\ldots,X_n$ to have *at least one* free occurrence in Φ. But even with that requirement it is quite controversial; e.g. it will be rejected by proponents of *intensionalism*, the view that necessarily coextensive properties (or relations or propositions) are identical.

One more principle is introduced on p. 20, namely that anything non-qualitative is necessarily non-qualitative.[2]

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[1] PTPI 8: 'it seems intuitively that any property that is constructed by certain operations out of purely qualitative properties must itself be purely qualitative'.

[2] '[T]he thisness or identity of a particular individual is nonqualitative either at all places, times, and possible worlds at which it occurs, or at none of them.'

3. Defining qualitativeness

Khamara: property F is qualitative := no object x and binary relation R are such that F = bearing R to x. More generally: n-ary relation R is qualitatve :- no object x and n+1-ary relation S are such that $R = (\lambda y_1 \ldots y_n . S y_1 \ldots y_b x)$.

- Worries about sufficiency arise in a coarse-grained setting: take x=Cleopatra and R = λyz.y loves Cleopatra and does not love z, One might think that λy.Ryx is qualitative on the grounds that it is the one and only necessarily instantiated property, and thus identical to *being self-identical* which is qualitative by Inheritance.
- Worries about necessity arise in a fine-grained setting. Surely if any property is non-qualitative, *being between Paris and London* is. But *between* is a *ternary* relation, and it's not uncontroversial that we can get this property by plugging any object in to the second argument of some *binary* relation.

Adams's more complicated definition.

Lewis's approach: define 'qualitative' in terms of 'fundamental' ('perfectly natural').

4. Claims considered by Adams

About *properties*, Adams considers (1), (3), and (4), to which it might be helpful to add (2):

(1) (= PII) Necessarily, no two things have the same qualitative properties.

(2) Necessarily, every property is coextensive with some qualitative property.

(3) Every property is necessarily coextensive with some qualitative property.

(4) Every property is qualitative.

- (4) obviously implies (3) and (2) obviously implies (1).
- □(3) obviously implies (2), but the implication from (3) itself to (2) is more controversial.
- (3) implies (4) given *Intensionalism*, the view that necessary coextensiveness suffices for identity of properties. Adams also points out (PTPI 13) that given (3) the denial of (4) would be hard to argue for.
- Here's an argument from (1) to (2). Let F be any property. Let C be the collection of all qualitative properties which are such that all of their instances are F, and let Q be *having some member of* C. Q is qualitative since every member of C is. Every Q object is F. Suppose for contradiction that x is F but not Q. Let D be the collection of all qualitative properties of x, and let R be *having every member of D*. R is qualitative since every member of D is. x has R. By (1), nothing distinct from x has R. So R is in C, and thus x has Q after all: contradiction.

About *worlds*, Adams discusses claim (5). But arguably, what he really means is (5′).

(5) No two possible worlds have the same qualitative properties

(5′) (= Anti-haecceitism) No two possible worlds are such that the same qualitative propositions are true at them.

- (5′) implies (5) given that whenever p is a qualitative proposition, *being a world at which p* is a qualitative property. But prima facie, (5) does not imply (5′). Suppose that there is a world w at which the same qualitative propositions that are true as are true at the actual world, but such that at w, Jupiter has the qualitative properties actually had by Saturn and Saturn has the qualitative properties actually had by Jupiter. Then one qualitative property of w is *being a world such that for some planet x: x is the largest planet in its solar system, but at w, x is not the largest planet in its solar system.* The actual world does not have this property.
- Adams says that believers in PII 'may be expected to suppose…that no two possible worlds are exactly alike in all qualitative respects'. I think he is right that it would be really weird to accept (1) but deny (5). By contrast, it is unclear how one could argue from (1) to (5′).
- Adams (PTPI 12) gives what he describes as an argument from from (1) and (5) to a special case of (3), the claim that every *haecceity* is necessarily equivalent to some qualitative property. I think in fact only works if we interpret it as an argument from (1) and (5′) to (3). […]

This turns out to be one of the many places where it's a good idea to look for ways of reformulating ideas expressed using 'possible world' in a way that doesn't use that expression. So in the vicinity of (5′), we can consider:

(6) Every true proposition is necessitated by some true qualitative proposition.

(7) Every possibly-true proposition is necessitated by some possibly-true qualitative proposition

(8) Every proposition is necessarily equivalent to some qualitative proposition

(9) Every proposition is qualitative.

- (9) obviously implies (8) which obviously implies (7) and (6).
- (7) implies (6) given the supplemental premise that there is a truth that necessitates every truth.
- (7) is equivalent to (5′) given the supplemental premise that for any possible world, there is a proposition that is true at that world but not at any other possible world.
- (8) implies (9) given Intensionalism.

- An argument that (7) implies (8): let p be a proposition. Let C be the collection of all possibly-true qualitative propositions that necessitate p, and let q be the proposition that *some member of* C *is true.* q is qualitative since it's "definable from" C. q necessitates p since each of its members does. And p must also necessitates q. For if it didn't, $p \wedge \neg q$ would be possible, and hence by (7) necessitated by some possibly-true, qualitative proposition r. But since r necessitates $p \wedge \neg q$, it necessitates p, and hence is a member of C, and hence necessitates q, and hence necessitates both q and $\neg q$, and hence isn't possible after all: contradiction.
- Here's an argument from (2) and (8) to (3) based on Adams's. Let F be some property. Let C be the collection of pairs Q,q, where Q is some qualitative property and q is some qualitative proposition necessarily equivalent to *F is coextensive with Q*. Let R be the property *being an x such that for some Q,q in C, q and Qx. R is qualitative since C is.* R necessitates F. And F necessitates R. For suppose that $\Diamond\exists x(Fx \wedge \neg Rx)$. Then by (2), $\Diamond\exists x\exists Q(Fx \wedge \neg Rx \wedge \mathrm{Qual}(Q) \wedge \forall y(Qy \leftrightarrow Fy))$. So by the (controversial!) principle that there couldn't be any "new" qualitative properties, $\exists Q(\mathrm{Qual}(Q) \wedge \Diamond\exists x(Fx \wedge \neg Rx$ $\forall y(Qy \leftrightarrow Fy))$. By (8), there is a qualitative proposition q necessarily equivalent to $\forall y(Qy \leftrightarrow Fy)$. So Q,q are in C, so $(\lambda y.Qy \wedge q)$ necessitates R, so $\neg\Diamond\exists x(Qx \wedge \neg Rx \wedge$ $\forall y(Qy \leftrightarrow Fy))$: contradiction.

5. Two ways of taking Black-type thought experiments

Ambitious:

> P1 Possibly, there is a sphere x and sphere y such that x and y have exactly the same qualitative properties, and x is at a distance from y,
>
> P2 Necessarily, for any x and y, if x is at a distance from y, then $x \neq y$.
>
> C PII is false

But P2 is hard to defend.

Modest:

> P Possibly, there is a sphere x and sphere y such that x and y have exactly the same qualitative properties, and x is at a distance from y.
>
> *C* PII is false.

6. Arguments from almost-indiscernibility

Simple argument: it is possible for there to be two almost-indiscernible spheres. But 'the possibility of there being two objects in a given spatiotemporal relation to each other is not affected by any slight changes in such features as the color or chemical composition of one or both objects' (17); so it is possible for there to be two indiscernible spheres.

Argument from the necessity of distinctness:

P1. $a \neq b$

P2. Possibly (Fa and Fb)

P3. Necessarily, if Fx and Fy, x and y have the same qualitative properties

P4. If $x \neq y$, then necessarily $x \neq y$

C: Possibly $\exists x \exists y$($x \neq y$ and x and y have the same qualitative properties)

- In Adam's instantiation, *a* and *b* are Adams and his almost-indiscernible twin, and F is *dreaming of a seven-horned dragon while living a life of such-and-such qualitative character*.

Argument from the temporal fixity of distinctness

Adams constructs his initial case in such a way that the slight asymmetry doesn't arise until late (27 years after the birth of the twins). As well as supporting P2, this suggests that he might be able to weaken P4 to the claim that 'The mutual distinctness of two individual persons already existing cannot depend on something that has not yet happened' (PTPI 18).

7. From not-PII to haecceitism

> Consider, again, a possible world w_1, in which there are two qualitatively indiscernible globes; call them Castor and Pollux. Being indiscernible, they have of course the same duration; in both of them have always existed and always will exist. But it seems perfectly possible, logically and metaphysically, that either or both of them cease to exist. Let w_2, then, be a possible world just like w_1 up to a certain time t at which in w_2. Castor ceases to exist while Pollux goes on forever; and let w_3 be a possible world just like w_1 except that in w_3 it is Pollux that ceases to exist at t while Castor goes on forever. (PTIT 22)

We can see what's going on here as an argument that if anti-haecceitism is true, then any two qualitatively indiscernible things are necessarily qualitatively indiscernible.

P1 (not-PII): it is possible for there to be two things with the same qualitative properties.

P2: if P1, it is possible for there to be two things x and y and a qualitative property F such that (i) x and y have the qualitative properties, and (ii) possibly (Fx and not Fy).

8. Arguments from not-PII to haecceitism